

SIH26236 — Master Build Spec

AI-Based Intelligent Food Packaging Material Recommendation System for Food Commodities Ministry of Food Processing Industries (MoFPI) · Theme: Agriculture, FoodTech & Rural Development · Category: Software

1. The Problem Statement — what MoFPI actually asked for

Inputs a solution must accept: commodity type, moisture content, oil/fat content, pH, respiration rate, desired shelf life, storage temperature, relative humidity, transport conditions, storage type (ambient/chilled/frozen).

Outputs a solution must produce: recommended packaging material (LDPE, HDPE, PET, metalized films, aluminium foil laminates, biodegradable films, breathable films) plus specifications — OTR, WVTR, film thickness, sealability, gas permeability, mechanical strength, MAP suitability.

Optional extras the PS itself invites: shelf-life prediction, sustainability analysis, cost optimization, QR-based traceability, eco-friendly packaging recommendations.

Important: the official brief does **not** mention regulatory compliance anywhere. That's our differentiator layer, not the literal ask — so the core AI/ML recommendation engine must be real and demonstrable on its own, with the compliance layer built on top.

2. Our Solution — the one-line version

A bilingual packaging recommendation engine that doesn't stop at "what material the science suggests" — it also tells the processor exactly which FSSAI Schedule their food falls under, which BIS IS code that material must legally meet, what their label must legally say, and whether they need CPCB-EPR registration. Built around real PMFME/ODOP commodities, in Hindi and English, offline-first.

Three layers: 1. **Core recommendation engine** — satisfies the literal PS ask (property-based material + spec recommendation) 2. **India Compliance Layer** — our actual differentiator (FSSAI + BIS + labelling + EPR + NABL, chained) 3. **Accessibility layer** — Hindi/English, voice input, offline-first, PMFME/ODOP-specific seed data

3. What Makes Us Unique — distinguishing factors

The strongest known competitor (PackSmart AI — ML recommendation, respiration/MAP science, shelf-life simulator, cost calculator, sustainability scoring, PDF reports, farmer/researcher modes) has **zero** mention of FSSAI, BIS, Legal Metrology, Hindi, or any Indian regulation anywhere. That is the one clean, verified gap in an otherwise sophisticated field — don't claim we're more advanced everywhere, claim we close the one gap nobody else has touched.

	Existing tools	Us
Core engine	ML + material science	Same, plus versioned compliance layer
Regulatory citation	None found anywhere in the field	FSSAI Schedule + BIS IS code + migration limit, per recommendation
Label compliance	Not covered	Auto-generated Labelling & Display checklist
Language	English only	Hindi + English, voice-enabled
Target framing	Generic global sustainability	PMFME/ODOP scheme beneficiary
Connectivity	Hosted web app only	Offline-first
Output	On-screen result	Downloadable, printable compliance sheet

Why this survives copying: a competitor can copy the headline (“add compliance”) in an afternoon, but correctly chaining Schedule IV (food category → material class) with Schedule I-III (material class → IS code), plus a separate labelling regulation, plus EPR, plus real PMFME commodities, plus bilingual + voice, all working live — that’s five compounding pieces of real research nobody replicates at this depth under the same time pressure.

4. Full Build Spec — point by point

4.1 Data layer (build this first, before any UI)

One JSON file or SQLite table, one row per commodity. Fields:

- name_en, name_hi
- category — fruit/vegetable, grain/millet, spice, dairy, dry snack, oil/fat product, etc.
- moisture_pct, fat_oil_pct, ph_range
- respiration_class — Very Low / Low / Moderate / High / Very High / Extremely High (Kader’s bands)
- storage_type — ambient / chilled / frozen
- shelf_life_target_days
- recommended_material — LDPE / HDPE / PET / BOPP / metalized film / aluminium foil laminate / biodegradable film
- otr_range, wvtr_range, thickness_um
- map_suitable (bool) + gas mix if yes
- fssai_schedule — Schedule I/II/III (material class) + Schedule IV (food-category mapping)
- bis_code — e.g. IS 10146
- migration_note — flag for plastics: 60 mg/kg or 10 mg/dm², tested per IS 9845
- epr_flag (bool) — true if plastic, plus category (rigid/flexible/multilayer/compostable)
- nabl_note — standard reminder text (conformity certificate required)
- labelling_checklist — one shared template across all commodities

4.2 Seed commodities (15-20, PMFME/ODOP-real, not generic)

Makhana (Bihar) · ragi/bajra/jowar millets · moringa (Tamil Nadu/Andhra) · turmeric (Telangana) · minor forest produce (tendu/mahua, tribal districts) · pickle · papad · jaggery · honey · dried

fruits · spice blends · ghee.

4.3 Reference data to seed OTR/WVTR (verify before final lock, but legitimate published ranges)

Material	OTR (cc/m ² /day, ~25µm, 23°C)	WVTR (g/m ² /day, ~25µm, 38°C/90%RH)	Barrier rating
LDPE	7,000-8,500	16-23	Poor
BOPP/OPP	1,000-2,500	3.9-6.2	Moderate
HDPE	—	4.7-7.8	Moderate
CPP	2,300-3,100	9.3-11	Moderate
EVOH	0.1-0.2	22-124	High (O ₂)
Aluminium foil laminate	~0	~0	Excellent

Cite test standards alongside: ASTM D3985/ISO 15105-2 (OTR), ASTM F1249/ISO 15106 (WVTR).

4.4 Screens — build in this order

Screen 1 — Commodity input - Text search + microphone icon (Web Speech API, lang="hi-IN" or "en-IN") - Language toggle top-right, persists through the whole flow - Autocomplete against seeded list only (no free text — breaks the demo if unseeded) - Optional property sliders (moisture, fat, pH, shelf life, storage type) pre-filled but overridable

Screen 2 — Recommendation result - Ranked material (1 primary + 1 alternative) - Specs as simple bars/icons ("Oxygen barrier: Moderate"), raw numbers behind an expandable "technical details" toggle - "Why this material?" — one line generated from the property values, not hardcoded per commodity

Screen 3 — India Compliance Layer (spend the most demo time here) - FSSAI Schedule + food category - BIS IS code + one-line scope - Migration limit flag if plastic - NABL certificate reminder - Labelling checklist (checkboxes) - CPCB-EPR flag if plastic, with category

Screen 4 — Compliance & Packaging Readiness Sheet - One-page bilingual summary, "Download PDF" button - Your single most demo-able artifact — rehearse clicking straight to it

Do not build: login/auth, dashboard, marketing homepage.

4.5 Recommendation engine — the honest version

Pick one, be able to explain it plainly under questioning: - **Rule-based scoring:** weighted score of candidate materials against moisture/fat/pH/respiration/shelf-life thresholds derived from seed data. Not "AI" — say so if asked. Fast, transparent, explainable in one sentence. - **Lightweight ML:** scikit-learn DecisionTreeClassifier or KNeighborsClassifier, trained on the seeded rows (expand to ~60-100 via small realistic jitter), predicting recommended_material. Report a real accuracy number, however unglamorous.

Never claim deep learning or unexplained "AI-powered" — someone will ask what the model actually is.

4.6 Bilingual + voice

- Hardcode an en/hi string dictionary (~40-60 strings) across all

- screens and the PDF — no translation engine
- Voice input: Web Speech API SpeechRecognition feeding the existing autocomplete

4.7 Tech stack

- Frontend: React or Flutter
- Backend: FastAPI (Python) — hosts both scoring/ML and compliance lookup
- Data: JSON/SQLite (no need for Postgres at this scale)
- PDF: reportlab or styled HTML → print-to-PDF

4.8 Explicitly cut, mention only as roadmap on the slide

- Live NABL/CPCB/FSSAI API or portal integration
- Multi-language beyond Hindi/English
- User accounts, saved history
- Vendor/supplier marketplace
- Ungraceful handling of unseeded commodities — have one clean fallback message ready instead of a crash

5. Regulatory reference sheet — verify every one of these before locking data

- Food Safety and Standards (Packaging) Regulations, 2018 — Schedule I (paper/board), Schedule II (metal/alloys), Schedule III (plastics), Schedule IV (food-category → material mapping)
- Polyethylene → **IS 10146** · Polystyrene → **IS 10142** · PET → **IS 12252**
- General wrapping paper → IS 6615 · Folding box board → IS 1776 · Aluminium foil → IS 8970 · Greaseproof paper → IS 6622
- Overall migration limit: 60 mg/kg or 10 mg/dm², tested per **IS 9845**
- Food Safety and Standards (Labelling and Display) Regulations, 2018 — FSSAI logo + 14-digit license number, net quantity, MRP, batch/lot number, mfg/best-before/expiry dates, veg/non-veg mark, ingredient list, consumer care details
- Plastic Waste Management Rules (as amended) — CPCB EPR framework, registration on the CPCB EPR portal, categories: rigid/flexible/multilayer/compostable
- NABL-accredited lab conformity certificate — required for food-contact packaging materials
- Kader, A.A. (2002), *Postharvest Technology of Horticultural Crops*, UC Davis, 3rd ed. — respiration rate classification
- PMFME Scheme, MoFPI — 2 lakh credit-linked beneficiaries achieved, ~11 lakh direct + indirect jobs generated, 44% women, ODOP approach (use these numbers, not “26 lakh”)

6. PPT — 5 slides, condensed reference

1. **PS + Solution + USP + Distinguishing Factors** — header strip (PS ID, title, ministry, theme), one-line pitch, 4-6 USP bullets, the comparison table from Section 3
2. **Technical Approach + Stack + Architecture + Workflow** — 5-stage pipeline: commodity input → AI recommendation engine → material + specs → India compliance layer → compliance sheet; name real components (React/Flutter, FastAPI, scikit-learn, JSON/SQLite, reportlab)
3. **Feasibility, Viability, Challenges + Mitigation** — technical feasibility (low-risk, proven components), named risk table (stale regulatory data → versioned dataset; paid BIS standards → cite

codes not full text; no smartphone → SMS/IVR or DRP-assisted; wrong advice → advisory disclaimer; small ML sample → honest accuracy)

4. **Impact + Benefits** — lead with 11 lakh jobs tied to PMFME, narrow to reduced non-compliant packaging + spoilage + easier formal licensing + early EPR awareness; one line on SDG 2/9/12
5. **References** — the full list in Section 5 above

Persona for the opening line: a makhana processor in Katihar or a moringa FPO in Tamil Nadu who took a PMFME loan, has no packaging engineer, and currently copies whatever pouch the local supplier stocks with no way to check if it's even legal.

Never name a specific competitor repo on the slide — keep the direct comparison loaded for verbal Q&A only.

7. Timeline — submission on the 29th, building today and tomorrow

When	Do
Today AM	Lock schema + name; start filling all 15-20 commodity rows; scaffold FastAPI + stub scoring function; build all 4 screens against dummy data; write PPT skeleton + jury-question answers
Today PM	Finish regulatory fields per commodity, cross-check every IS code; wire real scoring to first few commodities; swap dummy→real data as rows land; fill Slide 2 + 5
Tomorrow AM	Build Screen 3 + 4 (compliance layer + PDF) — highest-value screens; finish labelling checklist; finalize Slide 1, 3, 4
Tomorrow PM	Bilingual string pass; add voice input; full bug bash across every seeded commodity on the actual demo device
Tomorrow eve	Record video backup of a full run-through; read PPT aloud once with a timer; assign Q&A roles by topic
29th	No new features — verify every citation number one final time, test on the real presentation setup, submit with buffer before the deadline

8. Jury Q&A — script these answers now, word for word

- **“Who actually faces this problem today?”** → the named persona above, not the problem statement read back
- **“This already exists, why yours?”** → existing tools are science-first, English-only, built for a global audience; we're the only one chaining the same science to India's FSSAI Schedule + BIS + labelling + EPR requirements, in Hindi, seeded on real PMFME/ODOP commodities

- **“Why hasn’t anyone solved this yet?”** → the regulatory data is public but scattered across separate FSSAI PDFs and paid BIS documents; no consumer-facing packaging-AI product has bothered to structure it because their market doesn’t need Indian citations
- **“What’s your accuracy?”** → give the real number from your trained model, plus its known weak spot
- **“What happens when the model is wrong?”** → advisory decision-support tool, disclaimer on every output, recommend verification with a licensed consultant/NABL lab before commercial use